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Enhanced Performance of Two- and Three-Terminal Perovskite/Silicon Tandem Solar Panels Through Optimized Orientation and Tracking

The worldwide requirement for the most efficient and sustainable solar energy solutions has driven the development of **bifacial and tandem photovoltaic (PV) technologies**. The adoption of bifacial perovskite–silicon (PVK–Si) tandem solar cells has provided a new way to improve solar energy conversion efficiency. This work provides a comprehensive analysis of the energy yield potential of east–west (E/W) and north–south (N/S) oriented bifacial perovskite–silicon tandem solar panels in both single-axis tracking and fixed-tilt configurations. We focus on heterojunction with intrinsic thin layer (HIT), two-terminal tandem (2TT), and three-terminal tandem (3TT) configurations to evaluate their performance under varying ground albedo conditions. A simulation model based on MATLAB computes the annual and seasonal energy yields for these configurations. The results show that tracking panels consistently outperform their fixed-tilt counterparts. Seasonal analysis also demonstrates that E/W-oriented tracking panels perform exceptionally well in summer, whereas N/S-oriented tracking panels maintain a relatively stable output in both summer and fall. The optimized panel orientation significantly mitigates the skewed power output peaks observed in conventional configurations by strategically adjusting subcell exposure to incident sunlight. This approach results in a more balanced energy distribution throughout the day. These findings highlight the advantages of tracking bifacial tandem photovoltaic systems over conventional fixed-tilt installations, particularly for sites that require consistent power output and higher energy yields. [DOI: 10.1115/1.4071046]

Keywords: albedo, perovskite–silicon (PVK–Si) solar cells, photovoltaics (PV), simulation, tandem solar cells, efficiency, renewable, simulation, solar

1 Introduction

In order to meet the energy requirement of the expected global population of approximately 10B by 2050, renewable energy resources such as photovoltaics (PVs) will be much needed as they provide a very cost-effective and environment-friendly energy solution that is flexible to various land scales [1]. A remarkable reduction in the levelized cost of energy (LCOE) is made through the rapid progression in the developed technologies of solar cells and modules, consequently making PV is ever more worthwhile. Next-generation solar technologies, for instance, bifacial solar cells, are projected to reach up to 40% of the market share by 2028 [2]. Moreover, solar tracking has also been integrated into

new utility-scale PV systems of around 70% since 2015 [3]. Location specified benefits of combining solar tracking with bifacial solar cells highlighting the potential for achieving higher efficiency have already quantified, as a result of which tracking bifacial solar farms are likely to be extensively deployed globally [4]. To further minimize the LCOE, PV industry is developing next-generation solar cell technologies based on monofacial and bifacial perovskite–perovskite (PVK–PVK), perovskite–silicon (PVK–Si), and perovskite/organic (PVK/BHJ) tandem solar cells. Arguably, among all these, PVK–Si tandem cell is the finest choice because of potentially cheap integration of solution-processed perovskite (PVK) along with already commercially developed Silicon (Si) solar cell technologies. These efforts are motivated by the potential efficiency gain of tandem solar cells and the optimal bandgap alignment of perovskite with silicon-heterojunction (SHJ) cells [5,6]. The commercialization of two-terminal (2T) tandem solar cells has compelled studies on their outdoor performance, concentrating on energy yield potential for particularly bifacial modules in fixed-tilt and tracking systems. Theoretical study by Dupre et al. [7] has stated a significant gain

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of 20 with fixed-tilt standalone bifacial systems as compared to bifacial single-junction cells, while Schmager et al. [8] deployed monofacial 2T tandem (2TT) and predicted a 21% and 32% gain in energy yield potential using single and dual-axis tracking systems, respectively, as compared to fixed-tilt configuration. Field study conducted by Babics et al. [9] using a standalone bifacial 2T tandem with a single-axis tracking system reported a 55% gain in daily energy yield relative to a fixed-tilt configuration in desert conditions around the summer solstice.

A global simulation reported by Jahangir et al. [10] demonstrated that at an average albedo of 30%, three-terminal (3T) and four-terminal (4T) tandem solar cells offer approximately 5% and 23% gain in annual energy production in comparison to the 2T tandem and single-junction SHJ solar cells, respectively. The first bifacial 4T tandem solar introduced by Kim et al. [11] in the lab gained 30.5% efficiency under 1 sun with 0.2 sun rear irradiance, while bifacial minimodules of the same cell under 20–40% albedo condition at outdoor tests yield 20% higher output potential as compared to monofacial ones. Manshanden et al. [12] recently conducted an outdoor experiment revealing up to 26% efficiency gain at 10% albedo with 100 cm² bifacial 4TT minimodules over monofacial counterparts. Monolithic 4TT configuration experiences optical losses, contact shading, lateral transport losses, and parasitic absorption because of additional contacts [13]. Contrarily, the 3T configuration with optimal tailoring of the energy band gap of bifacial interdigitated back contact (IBC) bottom subcell gives advantages by integrating an extra back contact for carrier extraction, mitigating recombination losses from series-connected current-matching limitations which further enhance the output energy yield [14]. Gota et al. demonstrated that in different locations in the USA, monofacial 3T based on perovskite–silicon tandems can deliver up to 3–9% more output energy yield than monofacial 2T. Their studies also depict that a 4T tandem regardless of its relaxed design limitations in the context of perovskite thickness and energy bandgap fails to outperform the energy yield of optimized 2T due to parasitic optical losses [15]. Tockhorn et al. experimentally showed 17.1% efficiency with monofacial 3T perovskite/silicon tandem cell featuring IBC, while their simulation result showed a potential efficiency of 27%. Furthermore, studies also highlight the design capability of 3T tandems (3TTs) to enhance the energy yield by minimizing recombination losses by facilitating the current extraction [16]. Ross Rucker and Birnie investigated the effects of array design parameters such as module height, inter-row spacing, and racking strategies on the annual energy yield of vertical bifacial solar installations. However, these analyses focus on conventional bifacial systems without considering advanced tandem architectures or dynamic tracking and orientation optimization, as explored in this study [17]. The study by Venu Gopal et al. numerically investigated the optical performance of V trough solar concentrators using bifacial and trifacial absorbers with various bottom reflector geometries, showing that absorber type and reflector design significantly influence irradiance capture and optical efficiency. Their work provides valuable insights for designing concentrator PV systems, which can be further leveraged to evaluate farm-scale energy yield under realistic operating conditions in bifacial tandem solar farms [18]. Wei et al. demonstrated that image-based sun tracking can considerably improve the irradiance collection by specifically aligning PV panels with the solar trajectory. Their work emphasized the importance of dynamic orientation control in maximizing PV output [19]. Recent optimization studies by Lu and Hajimirza highlighted the importance of intelligently controlling panel orientation to maximize annual energy yield. A particle-based dust deposition model was used to show that the dust accumulation can substantially shift the optimal tracking angle of PV panels. This highlighted that the real environmental effects can reduce the efficiency of the conventional sun-tracking approaches, and model-based angle adjustment can maximize the irradiance collection [20]. Kazem et al. provided a comprehensive review of modern PV tracking technologies, including single and dual-

axis tracking systems, intelligent/algorithmic control strategies, structural designs, and performance metrics. The study highlighted how advanced tracking can improve irradiance capture, efficiency, and energy yield across climates, while also discussing emerging trends such as sensor-less control, artificial intelligence (AI) based optimization, and adaptive algorithms for diffuse-light conditions [21]. The study on the performance of PV tracking systems by Rodríguez-Gallegos et al. compared traditional sun-tracking with the best-orientation tracking, showing that orienting panels toward the direction of maximum instantaneous irradiance can improve energy yield, especially under diffuse or rapidly changing sky conditions [22]. However, their work focused on monofacial single-junction systems and did not investigate how such orientation strategies could influence bifacial or tandem (2T/3T) architectures, where rear-side irradiance, albedo, and multiterminal current extraction can introduce additional performance dynamics.

This work is the continuation of our previous work on design considerations of farm-level bifacial PVs. We have herein analyzed how vertical bifacial 2T/3T tandem solar farms perform under varying the irradiance and albedo condition. Along with this, it has also addressed how vertical bifacial 3T tandem configuration attenuates issues like current-matching in 2T with overall improvement in system performance. However, most performance analysis in the literature have only focused on monofacial tandem cells, and the energy yield potential of bifacial farms is largely unexplored under realistic conditions.

In this article, we extensively address the abovementioned gap by performing the annual energy yield potential analysis for tracking bifacial 2T and 3T tandem solar farm under realistic temperature varying condition. Additionally, toward this goal, we also developed a detailed simulation model to validate the influence of the orientation of panels (e.g., east–west or north–south) along with their impact on the energy yield of tracking bifacial 2T and 3T tandems and their constituent single-junction Si/SHJ cells.

2 Modeling Framework

Bifacial tandem solar farm comprising both vertical E/W and fixed-tilt N/S-oriented solar modules, with heterojunction with intrinsic thin layer (HIT), 2TT, and 3TT configuration subtypes

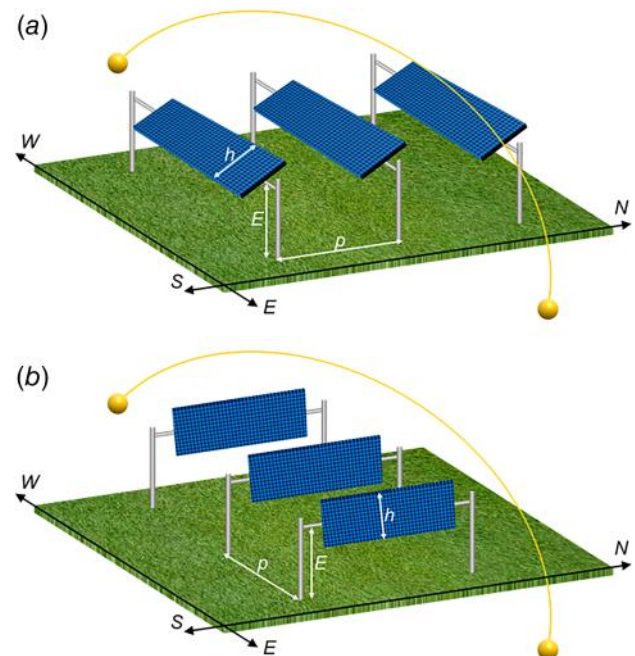


Fig. 1 Schematic diagrams of modeled PV farms: (a) tilted bi-N/S and (b) vertical bi-E/W

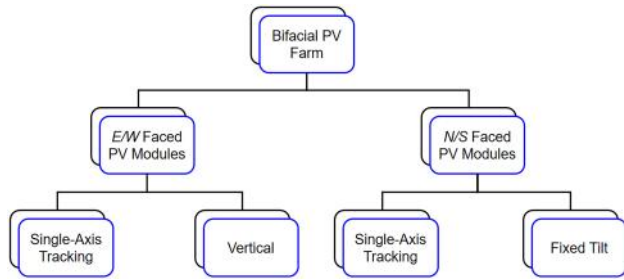


Fig. 2 Bifacial tandem solar farm with its subtypes of solar modules under each orientation

of solar modules for each orientation, is shown in Figs. 1 and 2, respectively.

For the simulation of the bifacial tandem solar farm, we have used a modeling framework that has been explained previously in Ref. [23]. Solar radiation composed of a bifacial tandem solar farm with respect to any location on the planet earth is computed using an **irradiance model**. For the calculation of the daily sun path elucidated by the sun's **azimuth** and **zenith** angles, the **PVLIB** toolbox from Sandia National Library is used for any particular location [24]. Thereafter, a **Haurwitz clear sky model** [25] is used for the computation of global horizontal irradiance (GHI), after which it is **scaled with the monthly average data from the NASA surface meteorology and solar energy database** [26]. By **Orgill and Hollands Perez's model** [27], beforehand computed GHI is decomposed into its two subcomponents, i.e., diffuse horizontal and direct normal irradiance.

Sun's azimuth angle (γ_M) relative to north, ground albedo (RA), module tilt angle (β) from the ground, panel's height (h) of 1 m, row-to-row spacing or pitch (p) of 2 m, and the elevation (E) of 1 m are used for defining PV array configuration in the farm. With these design attributes, the model calculates the **spatial**

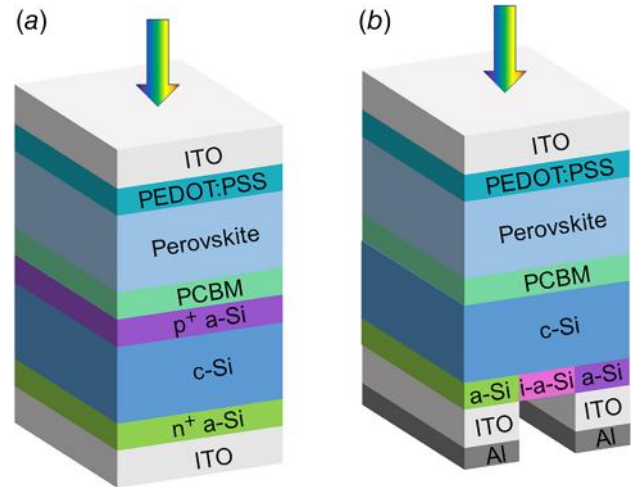


Fig. 3 Structure of (a) 2TT PVK/HIT and (b) 3TT PVK/HIT(IBC)

distribution of incident sunlight over the surface of modules and ground. A Two-dimensional-view factor approach that is explained in detail in our previous work [28] is applied for the calculation of output energy generated by albedo and diffused irradiance. A set of all these calculations is carried out at an **interval of one minute per day**. Moreover, to get maximum output power, we are considering the horizontal single-axis tracking system with both east–west and north–south orientations shown in Fig. 1. The module tilt angle (β) about the fixed horizontal axis is updated at each time-step for the extraction of maximized output power. **Silvaco ATLAS tool** simulates individual subcells for tandem PVK/Si solar cells, which are then combined at the module level with a VM (voltage matching) string ratio of 3/2 and cell area of 15.6 cm × 15.6 cm, and these

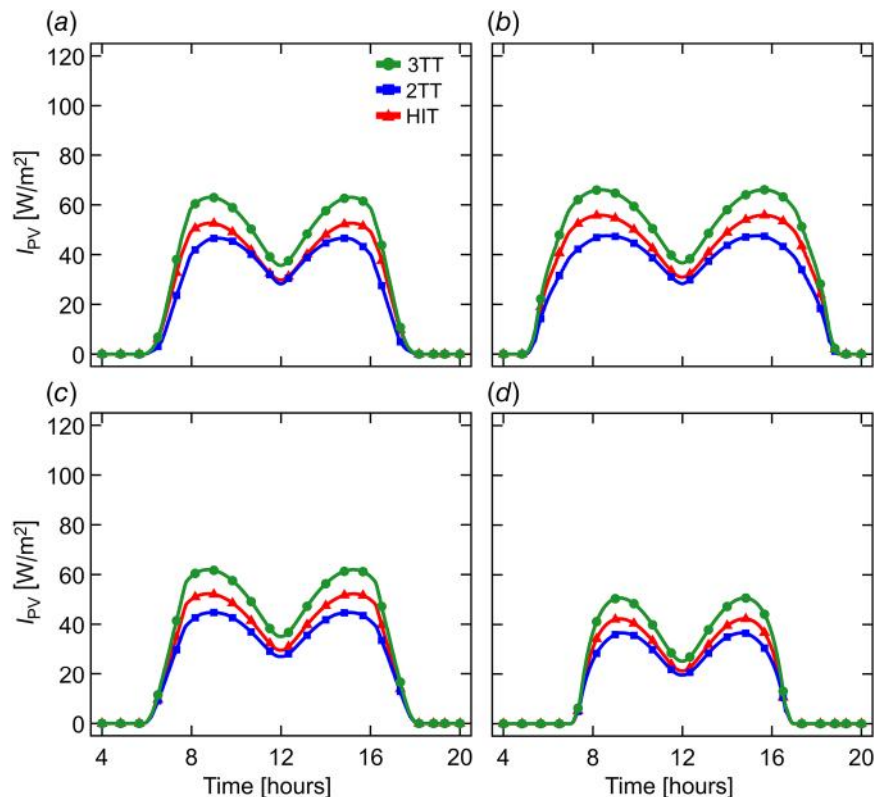


Fig. 4 Daily energy yield of vertical bi-E/W 3TT, 2TT, and HIT for various months for $p/h = 2$, $RA = 30\%$, and $E = 1$ m

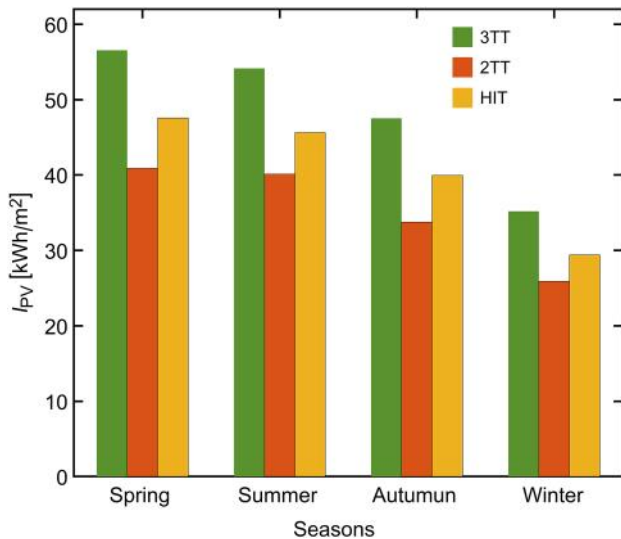


Fig. 5 Seasonal energy yield of vertical bi-E/W 3TT, 2TT, and HIT panels for $p/h = 2$, $RA = 30\%$ and $E = 1$ m

results are further processed to model the performance of bifacial PV arrays at the farm level [29].

At an individual level, the configuration of each subcell of single-junction HIT and tandem solar cell including both 2TT and 3TT has already been previously described in detail; here, we are summarizing its key points. The cell configuration of 2TT is PVK/s/HIT, and it is a series-connected configuration in which positive charge carriers (holes) extraction is carried out via a top contact, on the other hand, negative charge carriers (electrons) extraction is carried out via a back contact, while 3TT is PVK/r/nuIBC which is a reverse-connected configuration in which the

third contact on the back is meant for the collection of electrons from both subcells, and the subcells are connected with the same n-type doping at their common interface as shown in Fig. 3, respectively, with this nomenclature explained in detail [6] which actually shortens the descriptors of 2TT and 3TT cell configurations. The 3TT configuration ensures the maximum charge carriers collection, resulting in the implication of high efficiency for 3TT by evading current mismatching. In a 3TT tandem solar cell, the bottom IBC subcell incurs an inherent loss associated with albedo photocurrent due to the extra back contact shading as it blocks more reflected light in comparison to the conventional bottom HIT subcell in the 2TT cell which has only one back contact. However, this extra third back contact avoided the current mismatching constraints that are inherently present in the 2TT solar cell.

To reduce voltage mismatch in parallel and series interlacing of 3TT cells, connections with the help of VM strings are assumed, although this technique results in end losses of two out of 72 cells. Contrarily, there is no need for voltage matching in the case of 2TT modules, and therefore, they incur no end losses. The ideal VM ratio for 3TT comes out to be 1.579 based on VMPP of each subcell, but in order to keep the design practical and minimize end losses, a VM ratio of 1.5 provides a close match, though a VM ratio of 5/3 would even be a better match, but it turns out to incur double the end losses consequently making 3/2 more efficient overall.

Finally, we have also incorporated the effect of location-based temperature variation by self-consistently calculating the cell temperature T_{cell} based on hourly average ambient temperature data of the past 5 years obtained from NASA Power and temperature-dependent efficiency $\eta(T_{cell})$. For the calculation of the effective temperature coefficient TC^* of a 3/2 VM ratio PVK/Si tandem, the temperature coefficient of PVK and HIT was taken to be $-0.379\%/K$ and $-0.213\%/K$, respectively [30]. After that, with this computed TC^* , T_{amb} , and total incident light, we calculated the temperature corrected efficiency $\eta(T_{cell})$. After applying temperature corrections, the electrical model adjusts for partial

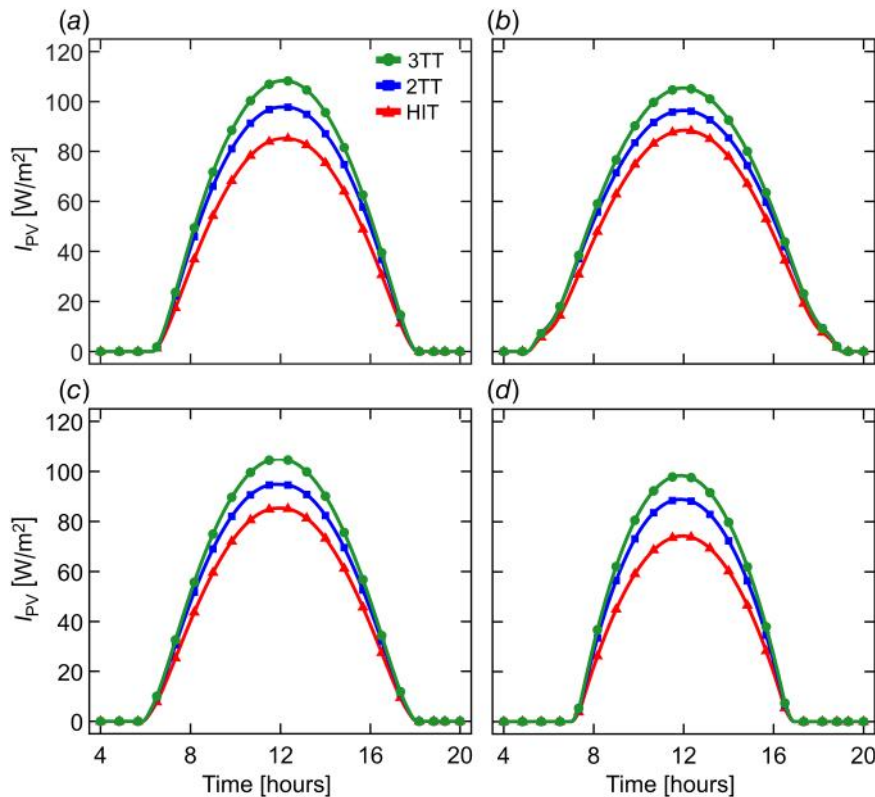


Fig. 6 Daily energy yield of fixed-tilt bi-N/S 3TT, 2TT, and HIT for various months for $p/h = 2$, $RA = 30\%$, and $E = 1$ m

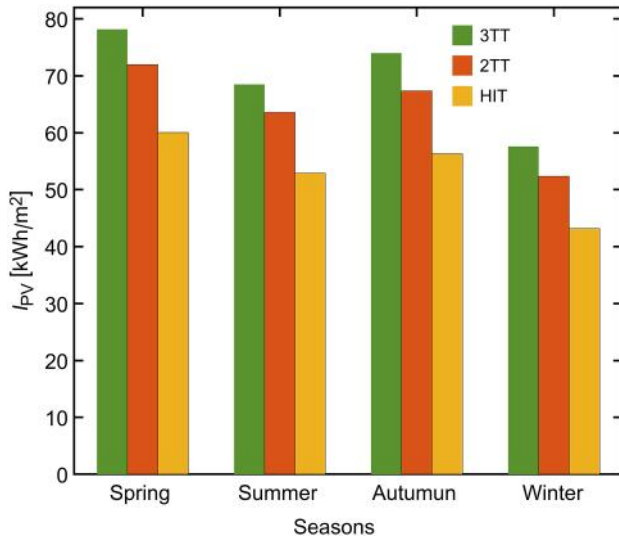


Fig. 7 Seasonal energy yield of fixed-tilt bi-N/S 3TT, 2TT, and HIT panels for $p/h = 2$, $RA = 30\%$, and $E = 1$ m

shading losses calculated in our previous work [31] and calculates the solar farm's power output at each step.

3 Results and Discussions

3.1 Energy Yield of E/W-Oriented Vertical Bifacial 3TT, 2TT, and HIT Solar Farms. The simulation results of daily energy output for HIT, 2TT, and 3TT are shown in Fig. 4 for different months. To elucidate the impact of the panel orientation, the issue of skewed morning and afternoon peaks in the daily energy

output profile of 2TT and 3TT as in our previous work [31]. Since the top PVK subcell for both 2TT and 3TT was facing east, it leads to a prominent peak in the morning and a slightly lower peak in the afternoon. We modified the panel orientation by positioning half of the panels with their top PVK subcell facing east and the other half with their bottom HIT subcell facing east, to achieve the balanced peaks, ensuring uniform power output in both morning and afternoon.

A similar trend of skewed morning and afternoon peaks in the daily energy output profile of HIT was observed too due to the bifaciality factor. However, by modifying the orientation in such that half of the HIT panels face their front side toward the east, and the other half have their back side facing east, the adjustment allows the panels to receive direct sunlight on the front side of HIT in the afternoon when the sun shifts westward, leading to balanced power output in both the morning and afternoon.

To obtain more realistic results, we incorporated the effect of location-specified temperature variations by self-consistently calculating the cell temperature and then temperature-dependent efficiency to assess the energy output as discussed in the modeling framework. It is observed that there is a slight decrement in yearly energy output of 3TT, 2TT, and HIT than in our previous findings. Meanwhile, our results support the same trend regarding the performance shown in Fig. 5, and the seasonal output is uppermost in spring, then tailed by summer and autumn, and lowermost in winter. Seasonal output energy shows that 3TT overtakes around 16% and 27% than HIT and 2TT, respectively. Overall yearly energy output yield of 3TT, 2TT, and HIT is 193.4 kWh/m^2 , 140.6 kWh/m^2 , and 162.5 kWh/m^2 respectively.

3.2 Energy Yield of N/S-Oriented Fixed Tilt Bifacial 3TT, 2TT, and HIT Solar Farms. Figure 6 shows the simulated results of daily energy output for HIT, 2TT, and 3TT for various months. Unlike E/W orientation, their daily energy output profile

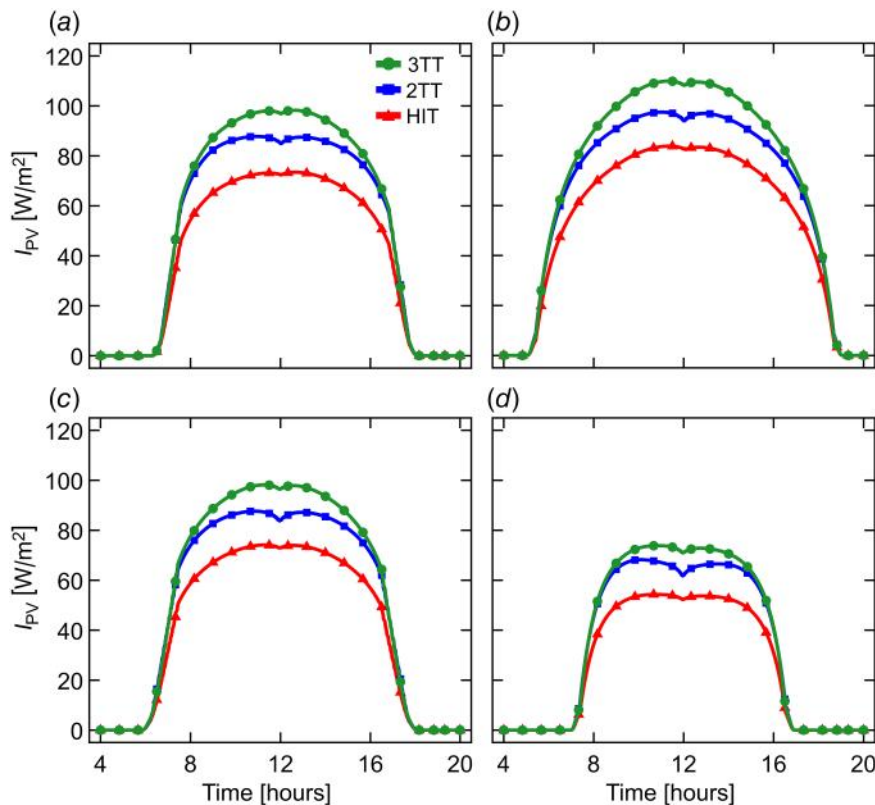


Fig. 8 Daily energy yield of tracking bi-E/W 3TT, 2TT, and HIT for various months for $p/h = 2$, $RA = 30\%$, and $E = 1$ m

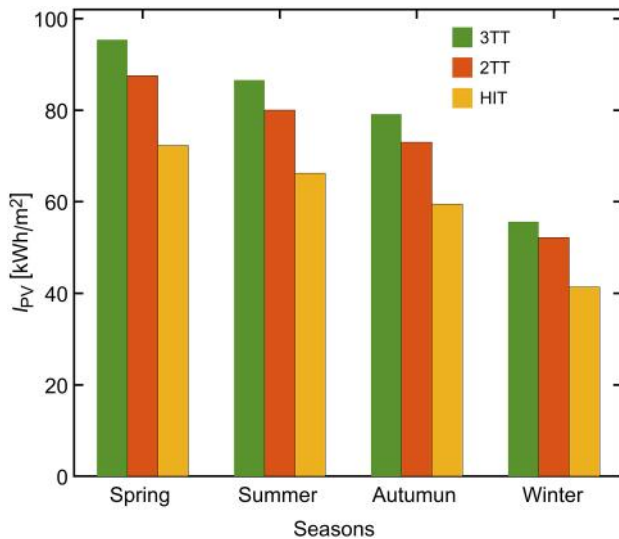


Fig. 9 Seasonal energy yield of tracking bi-E/W 3TT, 2TT, and HIT panels for $p/h = 2$, $RA = 30\%$, and $E = 1$ m

holds only one midday peak, and it is observed that the performance pattern follows a consistent trend of increasing energy output from HIT to 2TT and then 3TT. N/S orientation, 2TT configuration is uniformly receiving both direct and diffuse solar irradiance throughout the day thereby reducing the current mismatching factor that could limit its efficiency, and thus improve its performance as compared to HIT. The seasonal variation in the performance of 3TT, 2TT, and HIT shown in Fig. 7 highlights the impact of sun trajectory and seasonal variations of solar angle. Seasonal energy output is highest in spring, followed by autumn

and summer, and lowest in winter. In comparison to E/W orientation, the seasonal output of autumn surpasses summer because in autumn sun is usually lower in the sky and hence follows a more southern arc which results in N/S-oriented tilted solar panels to receive more direct sunlight, making them more effective in this season than E/W-oriented vertical bifacial solar panels. The overall yearly energy output yield of HIT, 2TT, and 3TT is 212.4 kWh/m^2 , 255.3 kWh/m^2 , and 278.2 kWh/m^2 , respectively.

3.3 Energy Yield of Tracking E/W-Oriented Bifacial 3TT, 2TT, and HIT Solar Panels. The significance of tracking solar panels in E/W-oriented bifacial configurations comes from the fact that these panels can change their orientation with respect to the sun's position throughout the day, thereby enabling them to capture solar energy all day long. This dynamic adjustment supports the system by making it possible to harvest greater energy than the fixed-tilt systems as they are only capable of capturing the sunlight at a static angle. Daily energy output yield of tracking E/W-oriented bifacial 3TT, 2TT, and HIT solar panels for various months is shown in Fig. 8. A slight drop in the intensity of energy output is observed during midday which is due to the effect of albedo. Other than that, tracking configuration also follows the same pattern regarding performance, i.e., 3TT outperforms 2TT while HIT underperforms both 2TT and 3TT. Seasonal output energy yield of tracking E/W vertical bifacial 3TT, 2TT, and HIT solar panels is shown in Fig. 9 which goes from highest to lowest in spring, summer, autumn, and winter, respectively, and it is quite visible that 3TT surpasses approximately 8% and 25% 2TT and HIT, respectively. The effect of tracking is more pronounced in 2TT and 3TT in comparison to fixed-mounted solar panel configurations. Yearly energy output yield of HIT, 2TT, and 3TT is 239.3 kWh/m^2 , 292.7 kWh/m^2 , and 316.7 kWh/m^2 , respectively.

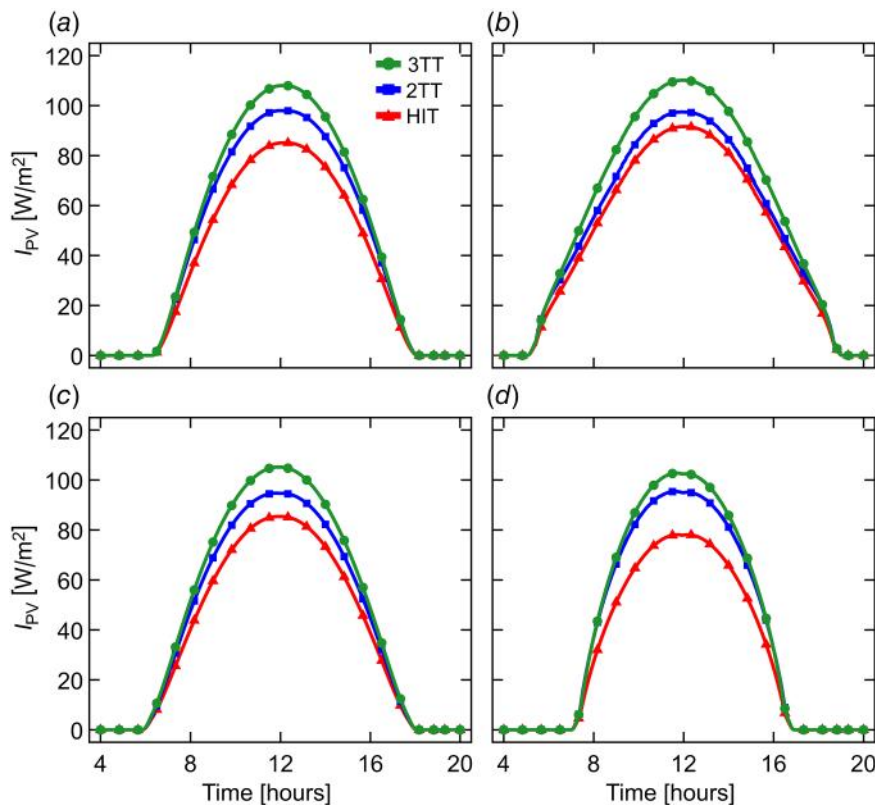


Fig. 10 Daily energy yield of tracking bi-N/S 3TT, 2TT, and HIT for various months for $p/h = 2$, $RA = 30\%$, and $E = 1$ m

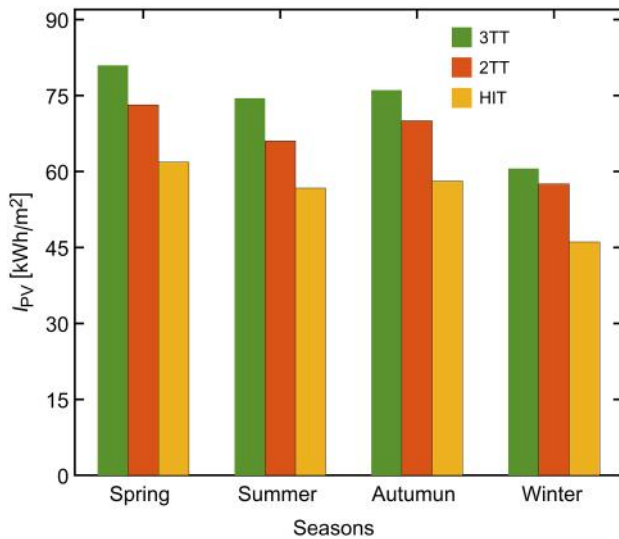


Fig. 11 Seasonal energy yield of tracking bi-N/S 3TT, 2TT, and HIT panels for $p/h = 2$, $RA = 30\%$, and $E = 1$ m

3.4 Energy Yield of Tracking N/S-Oriented Bifacial 3TT, 2TT, and HIT Solar Panels. The simulated results for the daily energy output yield of tracking N/S-oriented HIT, 2TT, and 3TT for various months are shown in Fig. 10. Unlike tracking the E/W-oriented configuration, the performance of tracking N/S-oriented configurations does not possess a significant advantage compared to its counterpart fixed-tilt configuration because N/S tracking panels are optimized for capturing sunlight primarily when the sun is directly overhead, thus limiting their efficiency during morning and evening hours. On the other hand, the performance trend of N/S tracking 3TT, 2TT, and HIT follows the same pattern as N/S fixed-tilt configuration such that 3TT outperforms 2TT and HIT underperforms both 3TT and 2TT.

Seasonal output energy yield of N/S tracking bifacial 3TT, 2TT, and HIT solar panels is shown in Fig. 11, which shows the highest yield in spring and the lowest in winter, while nearly consistent in summer and autumn in comparison to N/S fixed-tilt configurations because of their tracking mechanism which always ensures the panels align for capturing maximum sunlight. However, 3TT performs 24% better than HIT and 8% more than 2TT. The yearly energy output yield of N/S tracking HIT, 2TT, and 3TT is 222.7 kWh/m^2 , 266.7 kWh/m^2 , and 313.9 kWh/m^2 , respectively.

4 Conclusion

In this study, we have compared the performance of bifacial tandem perovskite/silicon solar PV farms in the vertical east/west faced and north/south faced fixed-tilted orientations with the single-axis tracking system in 2T and 3T IBC configurations. We modeled time-varying irradiance interception by the modules including direct, diffuse, and albedo components and the effect of location-based temperature to achieve high consistency. For 3T tandem IBC modules, we incorporated practical cell to module losses including shading of the extra back contact, voltage mismatch of the parallel strings, and end loss for the voltage-matched string. The following are the main conclusions of this article:

The performance of bifacial 2T tandem cells strongly depends on the time-varying solar spectrum and albedo, thus requiring careful optimization according to the use conditions to minimize current mismatch and the associated heating and reliability challenges.

3T IBC bifacial tandem configuration offers a flexible design as it avoids the current mismatch under various time-varying spectrum and albedo conditions.

Annual energy production of east/west-facing 3TT PVK/IBC vertical bifacial farm is 37.5% higher than the 2TT PVK/HIT vertical bifacial farm and 18% higher than single-junction HIT vertical bifacial farm.

Annual energy production of east/west-facing 3TT PVK/IBC tracking bifacial farm is 8.2% higher than the 2TT PVK/HIT tracking bifacial farm and 32.4% higher than single-junction HIT tracking bifacial farm.

East/west bifacial tracking farms outperform vertical bifacial farms by 63% (3TT), 108% (2TT), and 47% (HIT).

Annual energy production of north/south 3TT PVK/IBC fixed-tilt bifacial farm is 8.9% higher than the 2TT PVK/HIT fixed-tilt bifacial farm and 30.9% higher than single-junction HIT fixed-tilt bifacial farm.

Annual energy production of N/S 3TT PVK/IBC tracking bifacial farm is 9.4% higher than 2TT PVK/HIT tracking bifacial farm and 31.1% higher than single-junction HIT tracking bifacial farm.

N/S bifacial tracking farms outperform fixed-tilt farms by 5% (3TT), 4.5% (2TT), and 4.8% (HIT).

The results predict that the east/west-facing 3TT PVK/IBC tracking bifacial farm is the best configuration due to its significantly higher annual energy production compared to all other configurations as it outperforms 2TT and HIT tracking farms and provides substantial gains over vertical and fixed-tilt setups. The tracking capability and optimal east/west orientation maximize sunlight capture, making it the most efficient option, particularly for Lahore, Pakistan (latitude: 31.5° N , longitude: 74.3° E).

Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

Latin Symbols

- h = panel height (m)
- p = pitch (row-to-row spacing) (m)
- E = panel elevation above ground (m)
- T_{amb} = ambient temperature (K)
- T_{cell} = solar cell temperature (K)
- V_M = module voltage (V)
- V_{MPP} = voltage at maximum power point (V)
- DHI = diffuse horizontal irradiance (W/m^2)
- DNI = direct normal irradiance (W/m^2)
- GHI = global horizontal irradiance (W/m^2)
- TC^* = effective temperature coefficient ($\% \text{ K}^{-1}$)
- η = conversion efficiency (–)

Greek Symbols

- β = module tilt angle from ground (deg)
- γ_M = sun azimuth angle relative to north (deg)

Subscripts

- M = module
- amb = ambient condition
- cell = solar cell
- MPP = maximum power point

Abbreviations

- 2T = two-terminal
- 2TT = two-terminal tandem

3T = three-terminal
 3TT = three-terminal tandem
 4T = four-terminal
 4TT = four-terminal tandem
 HIT = heterojunction with intrinsic thin layer
 PV = photovoltaic
 PVK = perovskite
 RA = ground albedo
 Si = silicon

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